

Letters

TO THE EDITOR

The KETO CTA Study



The recent study by Soto-Mota et al¹ on plaque progression in lean, metabolically healthy individuals on a ketogenic diet with elevated low-density lipoprotein-cholesterol (LDL-C) offers timely insights. However, several methodological issues warrant consideration to contextualize the findings and assess their reliability.

Notably, the study preregistered the change in noncalcified plaque volume (Δ NCPV) as its primary outcome, yet this is only briefly shown in Figure 1 without proper description. Instead, the authors emphasize percent atheroma volume (PAV), a secondary endpoint. This shift from the prespecified analysis raises concerns of selective reporting. Additionally, the absence of a comparator group limits attribution of changes to the ketogenic diet.

Comparisons with other cohorts were purely descriptive, often involving very small subgroups (eg, $n = 17$), and lacked statistical testing or adjustment for confounders. Such indirect comparisons are difficult to interpret and should be presented with caution.

The validity of the regression models is also questionable. Despite claims of meeting assumptions, variables were reported using medians, suggesting non-normal distributions. Visual inspection of scatter plots shows clustering and no clear linear trends. Robust or nonparametric methods might have been more appropriate, and model diagnostics would improve transparency.

Crucially, Table 2 omits protein and total energy intake—key dietary factors influencing cardiometabolic outcomes. Moreover, models did not adjust for energy intake, body mass index, or body composition, potentially leading to residual confounding. Mediation analyses involving diet-related variables would have added clarity.

Although the use of Bayes Factors is a strength, the paper lacks a detailed explanation of the interpretive framework, and the joint reporting with P values is not clearly integrated.

The short follow-up (1 year) limits the ability to assess long-term cardiovascular impact. Prior

research suggests that 3 to 6 years may be required to observe meaningful plaque changes.² The focus on sensitivity analyses in the Central Illustration, despite discrepancies in reported sample size ($n = 57$ vs $n = 100$), may also mislead readers.

Importantly, the lack of association between apolipoprotein B or LDL-C and Δ NCPV in the short term does not negate decades of evidence linking elevated LDL-C with cardiovascular risk.³⁻⁵ These results should be viewed as hypothesis-generating, not practice-changing, especially in such a selective population.

Long-term, controlled studies in broader populations are needed to clarify the cardiovascular effects of ketogenic diets in hyper-responders.

Miguel López-Moreno, PhD^a

*José Francisco López-Gil, PhD^b

*One Health Research Group
Universidad de Las Américas
Quito, Ecuador

E-mail: josefranciscolopezgil@gmail.com

From the ^aUniversidad Francisco de Vitoria, Madrid, Spain; and the ^bUniversidad de La Américas, Quito, Ecuador.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

REFERENCES

1. Soto-Mota A, Norwitz NG, Manubolu VS, et al. Plaque begets plaque, ApoB does not: longitudinal data from the KETO-CTA trial. *JACC Adv*. 2025;4:101686. <https://doi.org/10.1016/j.jacadv.2025.101686>
2. Mendieta G, Pocock S, Mass V, et al. Determinants of progression and regression of subclinical atherosclerosis over 6 years. *J Am Coll Cardiol*. 2023;82:2069-2083.
3. Ference BA, Ginsberg HN, Graham I, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J*. 2017;38:2459-2472.
4. Borén J, John Chapman M, Krauss RM, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease: pathophysiological, genetic, and therapeutic insights: a consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J*. 2020;41:2313-2330.
5. Pedro-Botet J, Climent E, Benaiges D. LDL cholesterol as a causal agent of atherosclerosis. *Clin Investig Arterioscler*. 2024;36: Suppl 1:S3-S8.